

ME 300 Thermodynamics – Summer 2022

Exam 1

SOLUTIONS

(Part I) _____ (30 pts)

(Part II) _____ (40 pts)

(Part III) _____ (30 pts)

TOTAL _____

Directions

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3. Questions that require answers to be submitted to the online answers page on Canvas are written in *red italics* below as a way to remind you of what should be entered. All sketches and plots should be clearly drawn on your working paper and submitted with the upload.
4. If you have questions, please use the private chat function in Zoom to send a message directly to Dr. O'Connor so as not to disturb the other students. DO NOT UNMUTE AND ASK YOUR QUESTION ALOUD – this will be distracting to the other students.
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- I will not change any of my written answers after the exam but before I upload my work.
- I will not cheat in any way.

It is important to me to be a person of integrity and that means that all answers on this exam are my individual answers.
In the online answers, please type your full legal name to agree to the academic integrity pledge.

Part I: Concept Questions

1. (10 points) There are two tanks with the same volume and pressure – one with 10 moles of argon (MW=40 kg/kmol) and one with 10 moles of helium (MW=4 kg/kmol). *Similar to example in video M3.3*
- a) (5 points) Which tank has a higher temperature and why? *Answer this question in Canvas.*
- b) (5 points) Which one has a higher mass and why? *Answer this question in Canvas.*

Assume: Ar & He are noble gases $\rightarrow$ ideal gas +1

$$PV = NR_u T \rightarrow T = \frac{PV}{NR_u} \Rightarrow P, V, N \text{ are the same for both} +1$$

+2

R_u is a number

$$T_{Ar} = T_{He} +2$$

$$M = MW \times N \rightarrow MW_{Ar} > MW_{He}$$

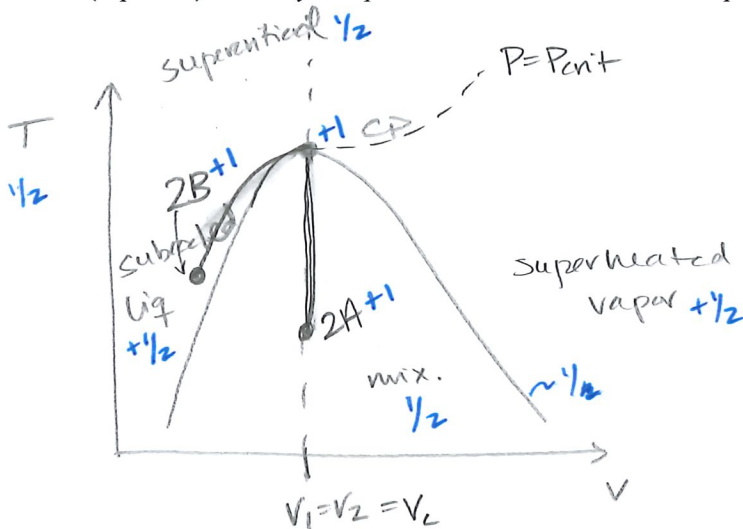
+2

$$M_{Ar} > M_{He} +2$$

2. (10 points) Two systems of water start at the critical point. One is cooled isochorically (Process A) and one is cooled isobarically (Process B). *Following diagram in video M4.2*

- a. (6 points) Draw these two processes on a T-v diagram, labeling the phases and the vapor dome. *Draw the figure below or on a separate piece of paper and upload it with your answers.*

- b. (4 points) Identify the phase of the water after each process. *Type your answer in the box in Canvas.*



Process A $\rightarrow$ mixture +2

Process B $\rightarrow$ subcooled liquid +2

3. (10 points) Helium is placed in a piston-cylinder system and is isothermally cooled from an initial volume V_1 to a final volume V_2 .

Following example in video 15.3

a) (2 points) Choose the correct statement about the change in internal energy of the helium. *Answer the multiple-choice question in Canvas.*

- A. $U_1 = U_2$ +2
- B. $U_1 > U_2$
- C. $U_1 < U_2$

b) (2 points) Choose the correct statement about V_1 and V_2 . *Answer the multiple-choice question in Canvas.*

- A. $V_1 = V_2$
- B. $V_1 > V_2$ +2
- C. $V_1 < V_2$

c) (2 points) What is the sign of work from State 1 to State 2? *Answer the multiple-choice question in Canvas.*

- A. Positive
- B. Negative +2

d) (4 points) Explain your answer to the previous question, "What is the sign of work from State 1 to State 2?" Use an equation to explain your answer. *Type your answer in the box in Canvas.*

Assume: He is ideal gas +1

a) for ideal gas $du = c_v dT \Rightarrow$ if $dT = 0$
 $du = 0$ +1

b) $PV = nRT$
 $\underbrace{PV}_{\text{const}} \uparrow \underbrace{T}_{\text{const}}$

cooling: $Q_2 < 0$

$\Rightarrow \Delta U = Q_2 - W_2$ +1
 $\rightarrow 0$

$Q_2 = W_2$ +1

$< 0 \quad < 0 \rightarrow V_2 < V_1$

c) $Q_2 = W_2$
 $< 0 \quad < 0$

Part II: Short Problems

1. (20 points) An inventor is testing out different processes for air in a piston-cylinder arrangement from the same initial state. In both processes, the mass-specific volume of the chamber increases from $v_1=5 \text{ m}^3/\text{kg}$ to $v_2=10 \text{ m}^3/\text{kg}$, but Process A is isobaric and Process B is isothermal. Answer the following questions about these two processes if the temperature at state 1 is $T_1=500 \text{ K}$. For air, $R=287 \text{ J/kg-K}$ and $c_v=714 \text{ J/kg-K}$. List your assumptions.

Following EE1

- (2 points) List the assumptions necessary to solve for the state points and work. *Type your answer in the box in Canvas.*
- (4 points) Draw both processes on the same P-v diagram and label the state points – State 1, State 2A (after Process A), and State 2B (after process B). *Draw the diagram below or on a separate page and upload your work with the rest of the exam – there is nothing to enter in Canvas.*
- (6 points) Calculate T_2 and P_2 for both processes. *Show your work below or on a separate page and enter these values in the Canvas quiz.*
- (3 points) Calculate the work of the isobaric process in kilojoules [kJ] if the mass inside the cylinder is 1 kg. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (5 points) Calculate the work of the isothermal process in kilojoules [kJ] if the mass inside the cylinder is 1 kg. *Show your work below or on a separate page and enter this value in the Canvas quiz.*

Given: $v_1 = 5 \text{ m}^3/\text{kg}$

$v_2 = 10 \text{ m}^3/\text{kg}$

$T_1 = 500 \text{ K}$

Process A: $P_1 = P_2$

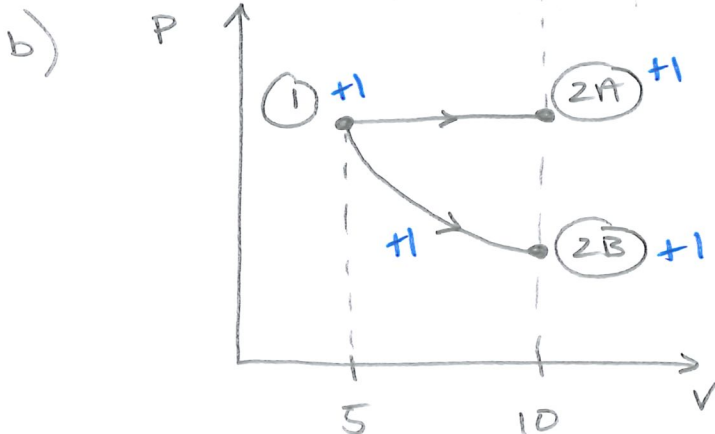
Process B: $T_1 = T_2$

a) Assume: air is ideal gas +1/2

$c_v = \text{const}$ +1/2

quasi-equ process +1/2

$\Delta u = \Delta pe = 0$ +1/2



c) State 1: $V_1 = 5 \text{ m}^3/\text{kg}$

$T_1 = 500 \text{ K}$

$P_1 V_1 = RT_1 \rightarrow P_1 = \frac{RT_1}{V_1} = \frac{(287)(500)}{5} = 28700 \text{ Pa} +1$

Process A: $\boxed{P_2 = P_1 = 28700 \text{ Pa}} +1$

$T_2 = \frac{P_2 V_2}{R} = \frac{(28700)(10)}{287} \rightarrow \boxed{T_2 = 1000 \text{ K}} +1$

Process B: $\boxed{T_2 = T_1 = 500 \text{ K}} +1$

$P_2 = \frac{RT_2}{V_2} = \frac{(287)(500)}{10} \rightarrow \boxed{P_2 = 14350 \text{ Pa}} +1$

d) $\int P dV = MP(V_2 - V_1) = (1)(28700)(10 - 5) \rightarrow \boxed{\bar{W}_2 = 143.5 \text{ kJ}} +1$

e) $\int P dV = \int \frac{MRT}{V} dV = MRT \ln(V_2/V_1) = (1)(287)(500) \ln(10/5) \rightarrow \boxed{\bar{W}_2 = 99.5 \text{ kJ}} +1$

2. (20 points) Water begins in a saturated liquid state at 450 K and is isothermally heated until it is a saturated vapor.

Following HW5 # 4

- a) (2 points) Write the assumptions necessary to solve for State 2 and the heat and work. *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- b) (2 points) What is the pressure of the water at State 1 in megapascal [MPa]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- c) (2 points) What is the pressure of the water at State 2 in megapascal [MPa]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- d) (6 points) What is the work done during this process in kilajoules [kJ] if the mass inside the cylinder is 1 kg? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- e) (8 points) What is the heat added to the system from State 1 to State 2 in kilojoules [kJ] if the mass inside the cylinder is 1 kg? *Show your work below or on a separate page and enter this value in the Canvas quiz.*

Given: $x_1 = 0$

$$T_1 = 450 \text{ K}$$

$$T_2 = T_1$$

$$x_2 = 1$$

a) Assume: S.C.S. +1

: quasi-ef. +1/2

: $\Delta K_e = \Delta P_e = 0$ +1/2

b) $P_1 = P_{\text{sat}} = 0.932209 \text{ MPa} - \text{Table D.1} +2$

c) $P_2 = P_{\text{sat}} = 0.932209 \text{ MPa} - \text{Table D.1} +2$

d) $\bar{W}_2 = \int P dV = MP(V_2 - V_1) = MP_{\text{sat}}(V_g - V_f)$
 $= (1)(0.932209 \times 10^6)(0.20781 - 0.0011232)$
 $\bar{W}_2 = 192.68 \text{ [kJ]} +2$

e) $\Delta U = Q_2 - \bar{W}_2 +2$

$Q_2 = \Delta U + \bar{W}_2 = M(u_g - u_f) + \bar{W}_2$
 $= (1)(2580.7 - 748.11) + 192.68 \sim \text{[kJ]}$

$Q_2 = 2025.27 \text{ [kJ]} +2$

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Part III: Analysis (30 points)

The story: A lot of large industrial processes have waste heat that comes off of the process that can't be used. In this problem, we're going to use that waste heat to run a cycle with water to produce some extra power for the factory. The cycle is listed below – assume that 10 kg of water is executing this cycle in a piston-cylinder device.

Following HW 4, #6

HW 5, #5

- Process 1-2: Isochoric heating
- Process 2-3: Isobaric expansion
- Process 3-4: Isochoric cooling
- Process 4-1: Isobaric compression

- a) (4 points) List the assumptions you'll need to solve for the state points as well as the first law quantities (heat, work, change in internal energy). *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- b) (12 points) Use the state points below to solve for the remainder of the states (P in [kPa], v in [m³/kg], and T in [K]). *Show your work below or on a separate page and enter this value in the Canvas quiz.*

State	Pressure [MPa]	Volume [m ³ /kg]	Temperature [K]
1	1.4	0.0012167	468.19
2	3	0.0012167	507
3	3	0.13995	920 K
4	1.4	0.13995	468.19

- c) (6 points) What is the net work of the cycle? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- d) (4 points) Draw this cycle on a T-v diagram, labeling all the state points. *Show your work below or on a separate page – you do not need to submit anything on the Canvas quiz for this part.*

Given: $M = 10 \text{ kg}$ +1

- a) Assume: water is S.C.S. +1
 : quasi-static process +1
 : $\Delta K.E. = \Delta P.E. = 0$ +1

- b) State 1: $P_1 = 1.4 \text{ MPa}$ +1
 $V_2 = V_1 = 0.0012167 \text{ m}^3/\text{kg}$ +1 $\rightarrow V_f < V_1 < V_g \Rightarrow \text{mixture}$ +1
 $T_1 = T_{\text{sat}} = 468.19 \text{ K}$ +1

State 2 : $P_2 = 3 \text{ MPa}$

$$V_2 = 0.0012167 \text{ m}^3/\text{kg} = V_f \rightarrow \text{sat. liq.} +1$$

$$\boxed{T_2 = T_{\text{sat}} = 507 \text{ K}} +1$$

State 3 : $\boxed{P_3 = P_2 = 3 \text{ MPa}} +1$

$$T_3 = 920 \text{ K} > T_{\text{sat}} \rightarrow \text{vapor} +1$$

$$\boxed{V_3 = 0.13995 \text{ m}^3/\text{kg}} +1 \text{ from Table D.3L}$$

State 4 : $\boxed{V_4 = V_3 = 0.13995 \text{ m}^3/\text{kg}} +1 \rightarrow V_f < V_4 < V_g \rightarrow \text{mix} +1$

$$\boxed{P_4 = P_1 = 1.4 \text{ MPa}} +1$$

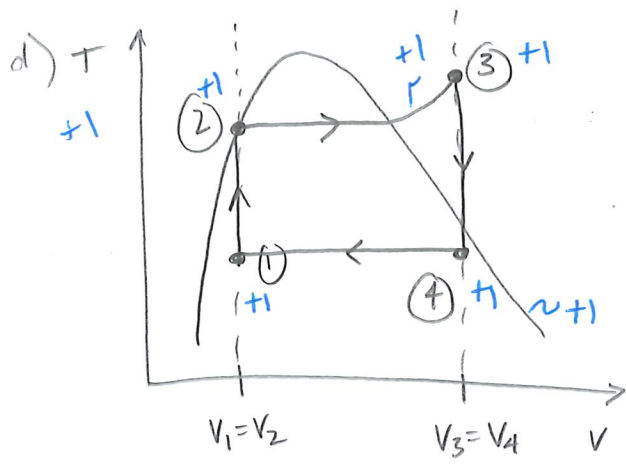
$$\boxed{T_4 = T_{\text{sat}} = 468.19 \text{ K}} +1$$

c) $\overset{+1}{\bar{W}_{\text{net}}} = \underset{\substack{\Delta V=0 +1}}{\bar{W}_2} + \underset{\substack{\Delta V=0 +1}}{2\bar{W}_3} + 3\bar{W}_4 + 4\bar{W}_1$

$$\rightarrow \underset{+1}{2\bar{W}_3} = \int P dV = MP(V_3 - V_2) = (10)(3 \times 10^6)(0.13995 - 0.0012167) = 4.162 \text{ MJ} +1$$

$$\rightarrow 4\bar{W}_1 = \int P dV = MP(V_1 - V_4) = (10)(1.4 \times 10^6)(0.0012167 - 0.13995) = -1.942 \text{ MJ} +1$$

$$\Rightarrow \boxed{\bar{W}_{\text{net}} = 2.22 \text{ MJ}} +1$$



ME 300 Thermodynamics – Summer 2022

Exam 1 – CONFLICT

SOLUTIONS

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Part I: Concept Questions

1. (10 points) There are two tanks with the same volume and pressure – one with 10 moles of neon (MW=20.2 kg/kmol) and one with 10 moles of krypton (MW=83.8 kg/kmol). *Similar to video M3.3*

- a) ~~(5)~~ ⁶ points) Which tank has a higher temperature and why? *Answer this question in Canvas.*
- b) ~~(5)~~ ⁴ points) Which one has a higher mass and why? *Answer this question in Canvas.*

Assume : Ne & Kr are noble gases $\rightarrow$ ideal gas +1

$$PV = N R_u T \rightarrow T = \frac{PV}{N R_u} \Rightarrow P, V, N \text{ are the same for both +1}$$

+2 R_u is a number

$$T_{Ne} = T_{Kr} \quad +2$$

$$M = MW \times N \rightarrow MW_{Kr} > MW_{Ne}$$

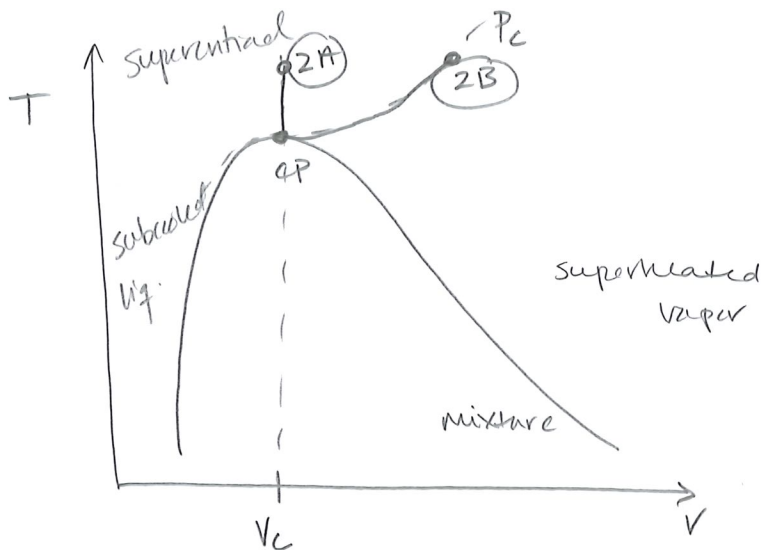
+2

$$M_{Kr} > M_{Ne} \quad +2$$

2. (10 points) Two systems of water start at the critical point. One is heated isochorically (Process A) and one is heated isobarically (Process B). *Following diagram in video M4.2*

- a. (6 points) Draw these two processes on a T-v diagram, labeling the phases and the vapor dome. *Draw the figure below or on a separate piece of paper and upload it with your answers.*

- b. (4 points) Identify the phase of the water after each process. *Type your answer in the box in Canvas.*



A) $\rightarrow$ supercritical fluid

B) $\rightarrow$ superheated vapor or
Supercritical $\rightarrow$ right on
the line

3. (10 points) Helium is placed in a piston-cylinder system and is isothermally heated from an initial volume V_1 to a final volume V_2 .

Following example in video M5.3

a) (2 points) Choose the correct statement about the change in internal energy of the helium. *Answer the multiple-choice question in Canvas.*

- A. $U_1 = U_2$ +2
- B. $U_1 > U_2$
- C. $U_1 < U_2$

b) (2 points) Choose the correct statement about V_1 and V_2 . *Answer the multiple-choice question in Canvas.*

- A. $V_1 = V_2$
- B. $V_1 > V_2$
- C. $V_1 < V_2$ +2

c) (2 points) What is the sign of work from State 1 to State 2? *Answer the multiple-choice question in Canvas.*

- A. Positive +2
- B. Negative

d) (4 points) Explain your answer to the previous question, "What is the sign of work from State 1 to State 2?" Use an equation to explain your answer. *Type your answer in the box in Canvas.*

Assume: He is ideal gas +1

: for ideal gas, $du = c_v dT \rightarrow$ if $dT = 0$
 $du = 0$ +1

'heating': $Q_2 > 0$

$$\Delta U = Q_2 - W_2 +1$$

$$Q_2 = W_2$$

$$> 0 \quad > 0 \quad \rightarrow \quad V_2 > V_1 +1$$

Part II: Short Problems

1. (20 points) An inventor is testing out different processes for air in a piston-cylinder arrangement from the same initial state. In both processes, the mass-specific volume of the chamber increases from $v_1=5 \text{ m}^3/\text{kg}$ to $v_2=10 \text{ m}^3/\text{kg}$, but Process A is isobaric and Process B is isothermal. Answer the following questions about these two processes if the temperature at state 1 is $T_1=500 \text{ K}$. For air, $R=287 \text{ J/kg-K}$ and $c_v=714 \text{ J/kg-K}$. List your assumptions.

Following EE1

- (2 points) List the assumptions necessary to solve for the state points and work. *Type your answer in the box in Canvas.*
- (4 points) Draw both processes on the same P-v diagram and label the state points – State 1, State 2A (after Process A), and State 2B (after process B). *Draw the diagram below or on a separate page and upload your work with the rest of the exam – there is nothing to enter in Canvas.*
- (6 points) Calculate T_2 and P_2 for both processes. *Show your work below or on a separate page and enter these values in the Canvas quiz.*
- (3 points) Calculate the work of the isobaric process in kilojoules [kJ] if the mass inside the cylinder is 1 kg. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (5 points) Calculate the work of the isothermal process in kilojoules [kJ] if the mass inside the cylinder is 1 kg. *Show your work below or on a separate page and enter this value in the Canvas quiz.*

See regular exam solution

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2. (20 points) Water begins in a saturated liquid state at 5.5 MPa and is isothermally heated until it is a saturated vapor.

Following HWS, #4

- a) (2 points) Write the assumptions necessary to solve for State 2 and the heat and work. *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- b) (2 points) What is the temperature of the water at State 1 in Kelvin [K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- c) (2 points) What is the temperature of the water at State 2 in Kelvin [K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- d) (6 points) What is the work done during this process in kilajoules [kJ] if the mass inside the cylinder is 1 kg? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- e) (8 points) What is the heat added to the system from State 1 to State 2 in kilojoules [kJ] if the mass inside the cylinder is 1 kg? *Show your work below or on a separate page and enter this value in the Canvas quiz.*

Given : $x_1 = 0$

$P_1 = 5.5 \text{ MPa}$

$T_2 = T_1$

$x_2 = 1$

a) Assume : S.C.S. +1

: quasi-st. +1/2

: $\Delta ke = \Delta pe = 0$ +1/2

b) $T = T_{\text{sat}} = 543.12 \text{ K} \rightarrow \text{Table D.2} +2$

c) $T = T_{\text{sat}} = 543.12 \text{ K} \rightarrow \text{Table D.2} +2$

d) $\int \bar{w}_2 = \int P dV = MP(v_2 - v_1) = MP(v_g - v_f)$
 $+1 \quad +1$
 $= (1)(5 \times 10^6)(0.035642 - 0.0013029)$
 $\boxed{\int \bar{w}_2 = 171.7 \text{ [kJ]}} +2$

e) $\Delta \bar{U} = \int \bar{Q}_2 - \int \bar{w}_2 +2$

$\int \bar{Q}_2 = \Delta \bar{U} + \int \bar{w}_2 = M(u_g - u_f) + \int \bar{w}_2$
 $+2$
 $= (1)(2593.7 - 1177.9) + 171.7$

$\boxed{\int \bar{Q}_2 = 1587.5 \text{ [kJ]}} +2$

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Part III: Analysis (30 points)

The story: A lot of large industrial processes have waste heat that comes off of the process that can't be used. In this problem, we're going to use that waste heat to run a cycle with water to produce some extra power for the factory. The cycle is listed below – assume that 10 kg of water is executing this cycle in a piston-cylinder device.

- Process 1-2: Isochoric heating
- Process 2-3: Isobaric expansion
- Process 3-4: Isochoric cooling
- Process 4-1: Isobaric compression

Follow HW4, #6
HWS, #5

- a) (4 points) List the assumptions you'll need to solve for the state points as well as the first law quantities (heat, work, change in internal energy). *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- b) (16 points) Use the state points below to solve for the remainder of the states (P in [kPa], v in [m³/kg], and T in [K]). *Show your work below or on a separate page and enter this value in the Canvas quiz.*

State	Pressure [MPa]	Volume [m ³ /kg]	Temperature [K]
1	1.4		
2	3	0.0012167	
3			920 K
4			

- c) (6 points) What is the net work of the cycle? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- d) (4 points) Draw this cycle on a T-v diagram, labeling all the state points. *Show your work below or on a separate page – you do not need to submit anything on the Canvas quiz for this part.*

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